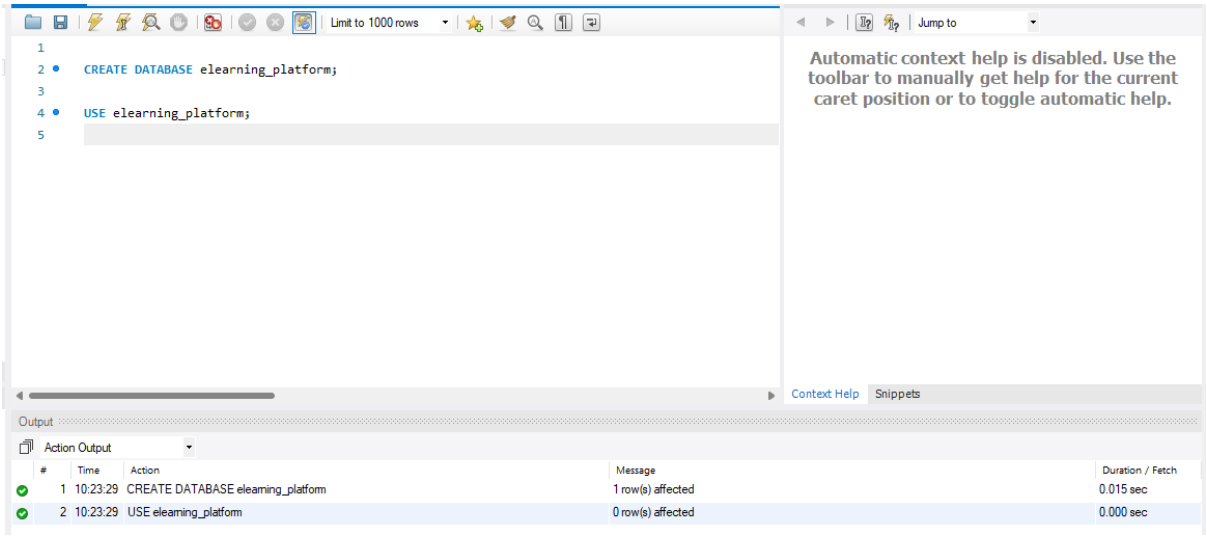
Module End Assignment:

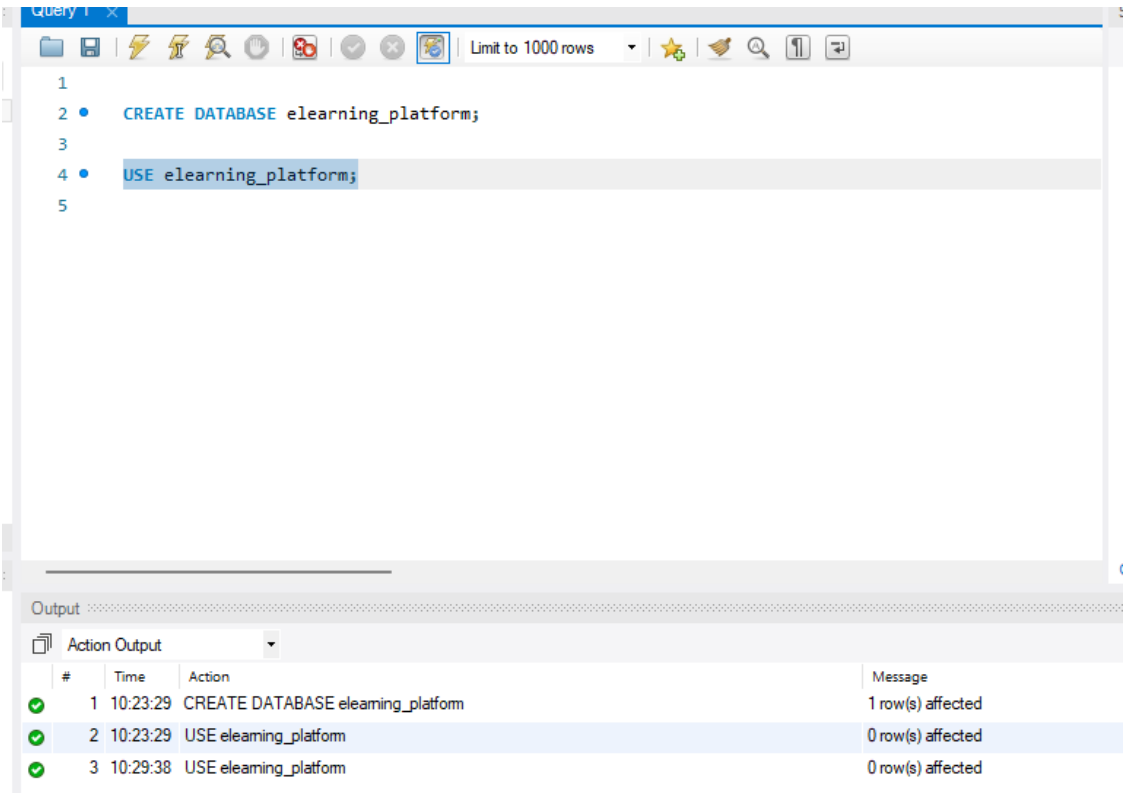
"Analyzing E-Learning Platform Purchases using MySQL"

Step 1 — Create database



Step 2: Create the three tables

- **Step 2.1 — Select your database**



➢ Step 2.2 — Create the learners table

```
5
6  CREATE TABLE learners (
7      learner_id INT PRIMARY KEY,
8      full_name VARCHAR(100) NOT NULL,
9      country VARCHAR(50) NOT NULL
0  );
1
```

put

Action Output

#	Time	Action	Message
1	10:23:29	CREATE DATABASE elearning_platform	1 row(s) affected
2	10:23:29	USE elearning_platform	0 row(s) affected
3	10:29:38	USE elearning_platform	0 row(s) affected
4	10:38:56	CREATE TABLE learners (learner_id INT PRIMARY KEY, full_name VARCHAR(100) NO...	0 row(s) affected

➢ Step 2.3 — Create the courses table

```
13 CREATE TABLE courses (
14     course_id INT PRIMARY KEY,
15     course_name VARCHAR(100) NOT NULL,
16     category VARCHAR(50) NOT NULL,
17     unit_price DECIMAL(10,2) NOT NULL
18 );
19
20
```

Output

Action Output

#	Time	Action	Message
1	10:23:29	CREATE DATABASE elearning_platform	1 row(s) affected
2	10:23:29	USE elearning_platform	0 row(s) affected
3	10:29:38	USE elearning_platform	0 row(s) affected
4	10:38:56	CREATE TABLE learners (learner_id INT PRIMARY KEY, full_name VARCHAR(100) NO...	0 row(s) affected
5	10:44:59	CREATE TABLE courses (course_id INT PRIMARY KEY, course_name VARCHAR(100) ...	0 row(s) affected

➢ Step 2.4 — Create the purchases table

```
21  CREATE TABLE purchases (
22      purchase_id INT PRIMARY KEY,
23      learner_id INT NOT NULL,
24      course_id INT NOT NULL,
25      quantity INT NOT NULL,
26      purchase_date DATE NOT NULL,
27
28      FOREIGN KEY (learner_id)
29          REFERENCES learners(learner_id),
30
31      FOREIGN KEY (course_id)
32          REFERENCES courses(course_id),
33
34      CHECK (quantity > 0)
```

Output

Action Output

#	Time	Action	Message
1	10:23:29	CREATE DATABASE elearning_platform	1 row(s) affected
2	10:23:29	USE elearning_platform	0 row(s) affected
3	10:29:38	USE elearning_platform	0 row(s) affected
4	10:38:56	CREATE TABLE learners (learner_id INT PRIMARY KEY, full_name VARCHAR(100) NO...	0 row(s) affected
5	10:44:59	CREATE TABLE courses (course_id INT PRIMARY KEY, course_name VARCHAR(100) ...	0 row(s) affected
6	10:47:23	CREATE TABLE purchases (purchase_id INT PRIMARY KEY, learner_id INT NOT NULL...	0 row(s) affected

➢ Step 2.5 — Verify that all tables were created

```
38  SHOW TABLES;
```

Result Grid | Filter Rows: | Export:

Tables_in_elearning_platform
courses
learners
purchases

Step 3: Insert sample data

➢ Step 3.1 — Insert data into learners

```
41  USE elearning_platform;
42
43  INSERT INTO learners (learner_id, full_name, country)
44  VALUES
45  (1, 'Arun Kumar', 'India'),
46  (2, 'Priya Sharma', 'India'),
47  (3, 'John Smith', 'USA'),
48  (4, 'Maria Garcia', 'Spain'),
49  (5, 'Chen Wei', 'Singapore');
50
51
52
```

Context Help

Output

Action Output

#	Time	Action	Message
4	10:38:56	CREATE TABLE learners (learner_id INT PRIMARY KEY, full_name VARCHAR(100) N...	0 row(s) affected
5	10:44:59	CREATE TABLE courses (course_id INT PRIMARY KEY, course_name VARCHAR(10...	0 row(s) affected
6	10:47:23	CREATE TABLE purchases (purchase_id INT PRIMARY KEY, learner_id INT NOT N...	0 row(s) affected
7	10:48:39	SHOW TABLES	3 row(s) returned
8	10:51:18	USE elearning_platform	0 row(s) affected
9	10:51:18	INSERT INTO learners (learner_id, full_name, country) VALUES (1, 'Arun Kumar', 'India'), (2, '...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

➢ Step 3.2 — Insert data into courses

```
INSERT INTO courses (course_id, course_name, category, unit_price)
VALUES
(101, 'Python Programming', 'Programming', 499.00),
(102, 'Data Analytics', 'Data Analytics', 599.00),
(103, 'Machine Learning', 'AI and ML', 799.00),
(104, 'Web Development', 'Programming', 449.00),
(105, 'Cloud Computing', 'Cloud Computing', 699.00);
```

Context Help Snip

ɔn Output

Time	Action	Message
10:44:59	CREATE TABLE courses (course_id INT PRIMARY KEY, course_name VARCHAR(10...	0 row(s) affected
10:47:23	CREATE TABLE purchases (purchase_id INT PRIMARY KEY, learner_id INT NOT N...	0 row(s) affected
10:48:39	SHOW TABLES	3 row(s) returned
10:51:18	USE elearning_platform	0 row(s) affected
10:51:18	INSERT INTO learners (learner_id, full_name, country) VALUES (1, 'Arun Kumar', 'India'), (2, '...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
10:55:00	INSERT INTO courses (course_id, course_name, category, unit_price) VALUES (101, 'Pytho...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

➢ Step 3.3 — Insert data into purchases

```
51
52 •  INSERT INTO courses (course_id, course_name, category, unit_price)
53    VALUES
54    (101, 'Python Programming', 'Programming', 499.00),
55    (102, 'Data Analytics', 'Data Analytics', 599.00),
56    (103, 'Machine Learning', 'AI and ML', 799.00),
57    (104, 'Web Development', 'Programming', 449.00),
58    (105, 'Cloud Computing', 'Cloud Computing', 699.00);
59
60
61
62
63
```

Context Help Sni

Output

Action Output

#	Time	Action	Message
6	10:47:23	CREATE TABLE purchases (purchase_id INT PRIMARY KEY, learner_id INT NOT N...	0 row(s) affected
7	10:48:39	SHOW TABLES	3 row(s) returned
8	10:51:18	USE elearning_platform	0 row(s) affected
9	10:51:18	INSERT INTO learners (learner_id, full_name, country) VALUES (1, 'Arun Kumar', 'India'), (2, '...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
10	10:55:00	INSERT INTO courses (course_id, course_name, category, unit_price) VALUES (101, 'Pytho...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
11	10:57:06	INSERT INTO purchases (purchase_id, learner_id, course_id, quantity, purchase_date) V...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0

➢ Step 3.4 — Verify your inserted data

```
60
61 •  SELECT * FROM learners;
62
63 •  SELECT * FROM courses;
64
65 •  SELECT * FROM purchases;
66
```

Result Grid | Filter Rows: | Edit:

learner_id	full_name	country
1	Arun Kumar	India
2	Priya Sharma	India
3	John Smith	USA
4	Maria Garcia	Spain
5	Chen Wei	Singapore
NULL	NULL	NULL

```
63 •  SELECT * FROM courses;
64
65 •  SELECT * FROM purchases;
66
```

Result Grid | Filter Rows: | Edit:

course_id	course_name	category	unit_price
101	Python Programming	Programming	499.00
102	Data Analytics	Data Analytics	599.00
103	Machine Learning	AI and ML	799.00
104	Web Development	Programming	449.00
105	Cloud Computing	Cloud Computing	699.00
NULL	NULL	NULL	NULL

```
65 •  SELECT * FROM purchases;
66
```

Result Grid | Filter Rows: | Edit: | Expor

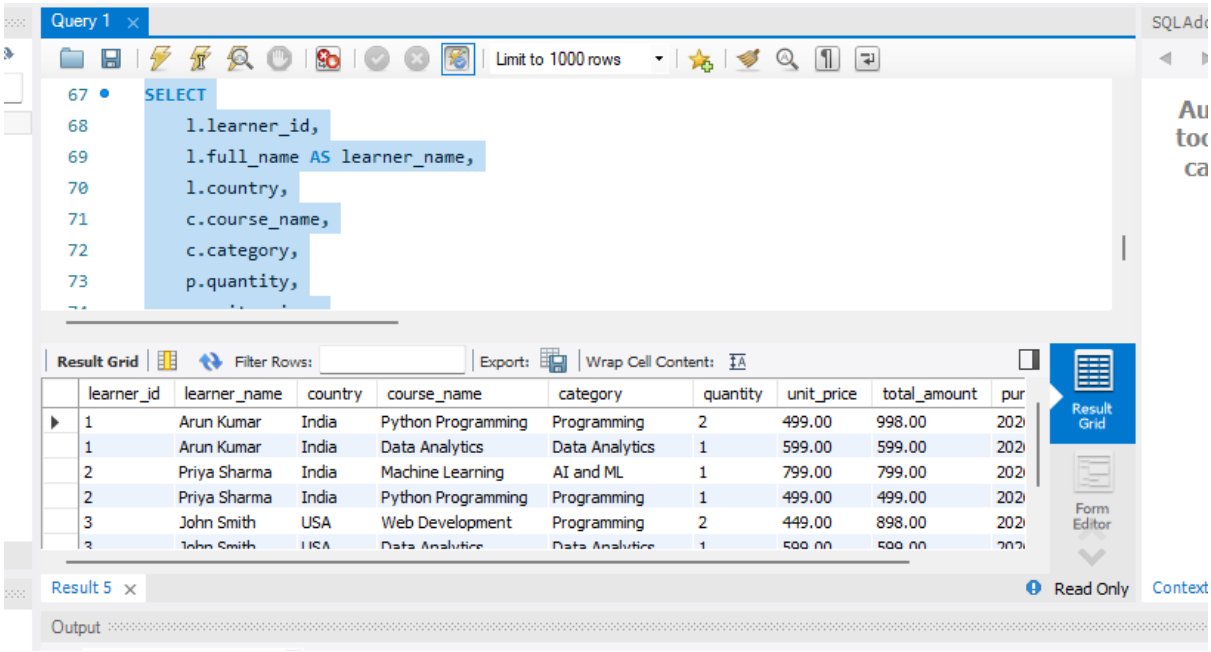
purchase_id	learner_id	course_id	quantity	purchase_date
1001	1	101	2	2026-10-01
1002	1	102	1	2026-10-02
1003	2	103	1	2026-10-03
1004	2	101	1	2026-10-04
1005	3	104	2	2026-10-05
1006	3	102	1	2026-10-06
1007	4	105	1	2026-10-07

Step 4: Data Exploration Using JOINs

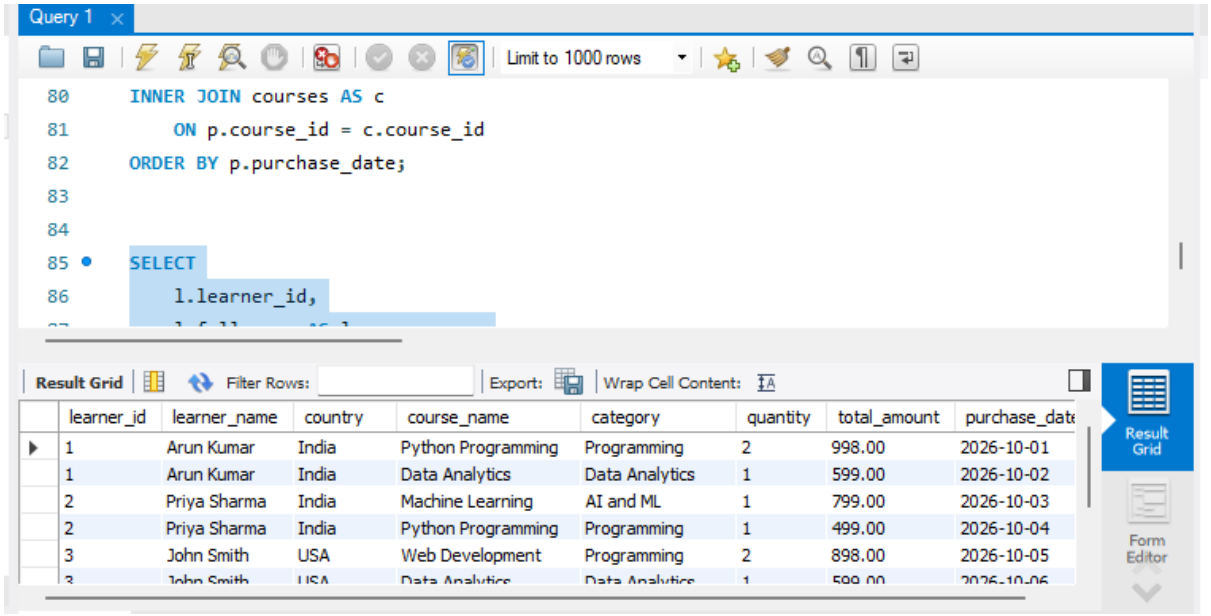
➢ Step 4.1 — Understand how the tables connect

The purchases table connects the learners and courses tables. We use learner_id and course_id to match the corresponding records.

➢ Step 4.2 — INNER JOIN



➢ Step 4.3 — LEFT JOIN



➢ Step 4.4 — RIGHT JOIN

```
    p.purchase_date
FROM purchases AS p
RIGHT JOIN courses AS c
    ON p.course_id = c.course_id
LEFT JOIN learners AS l
    ON p.learner_id = l.learner_id
ORDER BY c.course_id, p.purchase_date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_id	course_name	category	learner_name	quantity	total_amount	purchase_date
101	Python Programming	Programming	Arun Kumar	2	998.00	2026-10-01
101	Python Programming	Programming	Priya Sharma	1	499.00	2026-10-04
102	Data Analytics	Data Analytics	Arun Kumar	1	599.00	2026-10-02
102	Data Analytics	Data Analytics	John Smith	1	599.00	2026-10-06
103	Machine Learning	AI and ML	Priya Sharma	1	799.00	2026-10-03
103	Machine Learning	AI and ML	Maria Garcia	2	1,598.00	2026-10-08
104	Web Development	Programming	John Smith	2	898.00	2026-10-05

Result 7

Step 5: Analytical Queries Q1–Q5.

Q1. Display each learner's total spending along with their country

```
SELECT
    l.learner_id,
    l.full_name AS learner_name,
    l.country,
    FORMAT(SUM(p.quantity * c.unit_price), 2) AS total_spent
FROM learners AS l
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

learner_id	learner_name	country	total_spent
4	Maria Garcia	Spain	2,297.00
1	Arun Kumar	India	1,597.00
3	John Smith	USA	1,497.00
2	Priya Sharma	India	1,298.00

Q2. Find the top 3 most purchased courses

```
135 •  SELECT
136        c.course_id,
137        c.course_name,
138        SUM(p.quantity) AS total_quantity_sold
139    FROM courses AS c
140    INNER JOIN purchases AS p
141        ON c.course_id = p.course_id
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_id	course_name	total_quantity_sold
103	Machine Learning	3
101	Python Programming	3
102	Data Analytics	2

Q3. Show each category's total revenue and unique learners

```
149
150
151 •  SELECT
152        c.category,
153        FORMAT(SUM(p.quantity * c.unit_price), 2) AS total_revenue,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_revenue	unique_learners
AI and ML	2,397.00	2
Programming	2,395.00	3
Data Analytics	1,198.00	2
Cloud Computing	699.00	1

Toggle wrapping of cell contents

Q4. List learners who purchased from more than one category

```
163 •  SELECT
164        l.learner_id,
165        l.full_name AS learner_name,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

learner_id	learner_name	categories_purchased
1	Arun Kumar	2
3	John Smith	2
4	Maria Garcia	2
2	Priya Sharma	2

Q5. Identify courses that have never been purchased

```
179 •  SELECT
180        c.course_id,
181        c.course_name,
182        c.category,
183        FORMAT(c.unit_price, 2) AS unit_price
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_id	course_name	category	unit_price